

Curriculum vitae

Amir Yehudayoff

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Academic degrees

2008 Ph.D., Faculty of Mathematics and Computer Science, Weizmann Institute of Science

2005 M.Sc., Faculty of Mathematics and Computer Science, Weizmann Institute of Science

2003 B.Sc., Computer Science and Physics with a division in Mathematics, School of Computer Science, The Hebrew University of Jerusalem

Academic appointments

2023–present Professor, Department of Computer Science, The University of Copenhagen

2020–present Professor, Department of Mathematics, Technion–IIT

2015–2020 Associate Professor, Department of Mathematics, Technion–IIT

2018–2019 Visiting Associate Professor, EECS, UC Berkeley

2010–2015 Assistant Professor, Department of Mathematics, Technion–IIT

2008–2010 Member, School of Mathematics, Institute for Advanced Study, Princeton

Fellowships, awards and honors

2024 DNR Chair, Danish National Research Foundation

2020 The Kurt Mahler Prize for Excellence in Mathematics, Technion–IIT

2016 The Cooper Prize for Excellence in Scientific Research, Technion–IIT

2014 The Krill Prize for Excellence in Scientific Research, The Wolf Foundation

2013 The Kurt Mahler Prize for Excellence in Mathematics, Technion–IIT

2011 Horev Fellowship, Leaders in Science and Technology program, Technion–IIT

2011 Allon Fellowship for Outstanding Young Researchers

2008 Best Paper Award, Computational Complexity Conference

2006 Eshkol Fellowship, The Israeli Ministry of Science

Public professional activities

2021–present Member of the Minerva Fellowship Committee

2020–present Associate Editor in Journal of Computer and System Sciences

2018 Organizer of Lower Bounds in Computational Complexity semester, Simons Institute for the Theory of Computing, Berkeley CA

2016–2018 Secretary of the Israel Mathematical Union

Scientific committees of conferences

Senior program committee International Conference on Algorithmic Learning Theory 2027

Area chair NeurIPS 2026

IEEE Symposium on the Foundations of Computer Science 2021, 2017

Annual Conference on Learning Theory 2012, 2024

Computer Science in Russia 2021

Innovations in Theoretical Computer Science 2012

Computational Complexity Conference 2020, 2016, 2012

RANDOM 2013

Teaching experience

Many courses in a wide variety of topics (sorted alphabetically)

Analytic methods in combinatorics, graduate (designed course)

Applications of information theory, graduate (designed course)

Basic notions in mathematics, undergraduate

Calculus 1, undergraduate; Calculus 2, undergraduate

Combinatorial algorithms, undergraduate

Combinatorics, undergraduate

Communication complexity, graduate (designed course)

Computability and computational complexity, undergraduate

Data structures and algorithms, undergraduate

Introduction to probability, undergraduate

Mathematics for architects, undergraduate

Mathematics for life sciences, undergraduate

Mathematical aspects of learning theory, graduate (designed course)

Set theory, undergraduate

Students

Shay Moran, Ph.D.
Ido Nachum, Ph.D.
Daniel Carmon, Ph.D.
Daniel Avraham, M.Sc.
Jonathan Shafer, M.Sc.
Krishna Manoorkar, M.Sc.
Shachar Shabelman, M.Sc.

Publications

(authors are alphabetically sorted)

Book

A. Rao and A. Yehudayoff. Communication complexity and applications. Cambridge University Press, 2020

Journal papers

(some appeared in conferences; info in brackets)

1. I. Benjamini, A. Yadin and A. Yehudayoff. Random graph-homomorphisms and logarithmic degree. *Electronic Journal of Probability* 12, pages 926–950, 2007
2. R. Raz, A. Shpilka and A. Yehudayoff. A lower bound for the size of syntactically multilinear arithmetic circuits. *SICOMP* 38 (4), pages 1624–1647, 2008 (FOCS 2007)
3. R. Raz and A. Yehudayoff. Balancing syntactically multilinear arithmetic circuits. *Computational Complexity* 17 (4), pages 515–535, 2008
4. R. Gradwohl and A. Yehudayoff. t-wise independence with local dependencies. *Information Processing Letters* 106, pages 208–212, 2008
5. Z. Dvir, A. Shpilka and A. Yehudayoff. Hardness-randomness tradeoffs for bounded depth arithmetic circuits. *SICOMP* 39 (4), pages 1279–1293, 2009 (STOC 2008)
6. R. Peled, A. Yadin and A. Yehudayoff. The maximal probability that k-wise independent bits are all 1. *Random Structures & Algorithms* 38 (4), pages 502–525, 2009
7. R. Raz and A. Yehudayoff. Lower bounds and separations for constant depth multilinear circuits. *Computational Complexity* 18 (2), pages 171–207, 2009 (best paper award CCC 2008)
8. R. Gradwohl, O. Reingold, A. Yadin and A. Yehudayoff. The player’s effect. *Mathematics of Operations Research* 34 (4), pages 971–980, 2009

9. P. Hrubeš and A. Yehudayoff. Monotone separations for constant degree polynomials. *Information Processing Letters* 110, pages 1–3, 2009
10. I. Benjamini, G. Kozma, A. Yadin and A. Yehudayoff. Entropy of random walk range. *Annales de l’Institut Henri Poincare* 6 (4), pages 1080–1092, 2010
11. R. Raz and A. Yehudayoff. Multilinear formulas, maximal-partition discrepancy and mixed-sources extractors. *Journal of Computer and System Sciences* 77 (1), pages 167–190, 2011 (FOCS 2008)
12. A. Yadin and A. Yehudayoff. Loop-erased random walk and poisson kernel on planar graphs. *Annals of Probability* 39 (4), pages 1243–1285, 2011
13. A. Yehudayoff. Affine extractors over prime fields. *Combinatorica* 31 (2), pages 245–256, 2011
14. P. Hrubeš and A. Yehudayoff. Homogeneous formulas and symmetric polynomials. *Computational Complexity* 20 (3), pages 559–578, 2011
15. P. Hrubeš and A. Yehudayoff. Algebraic complexity in ring extensions. *Theory of Computing* 7, pages 119–129, 2011
16. P. Hrubeš, A. Wigderson and A. Yehudayoff. Noncommutative circuits and the sum-of-squares problem. *Journal of the AMS* 24, pages 871–898, 2011 (STOC 2010)
17. A. Yehudayoff. Linear cover time for trees is exponentially unlikely. *Chicago Journal on Theoretical Computer Science*, Volume 2012, Article 3, 2012
18. B. Barak, Z. Dvir, A. Wigderson and A. Yehudayoff. Fractional Sylvester-Gallai theorems. *PNAS* 110 (48), pages 19213–19219, 2013 (STOC 2011)
19. A. Bogdanov, Z. Dvir, E. Verbin and A. Yehudayoff. Pseudorandomness for width 2 branching programs. *Theory of Computing* 9, pages 283–293, 2013
20. J. Bourgain and A. Yehudayoff. Expansion in $SL_2(\mathbb{R})$ and monotone expanders. *Geometric and Functional Analysis* 23 (1), pages 1–41, 2013 (STOC 2012)
21. R. Peled, W. Samotij and A. Yehudayoff. Lipschitz functions on expanders are typically flat. *Combinatorics, Probability & Computing* 22 (4), pages 566–591, 2013
22. P. Hrubeš, A. Wigderson and A. Yehudayoff. An asymptotic bound on integer sum of squares. *Canadian Mathematical Bulletin* 56, pages 70–79, 2013
23. R. Peled, W. Samotij and A. Yehudayoff. Lipschitz functions on trees are typically flat. *Electronic Journal of Probability* 18 (55), pages 1–9, 2013
24. H. Duminil-Copin, C. Lucas, A. Yadin and A. Yehudayoff. Containing internal diffusion limited aggregation. *Electronic Journal of Probability* 18 (50), pages 1–8, 2013
25. M. Braverman, A. Rao, R. Raz and A. Yehudayoff. Pseudorandom generators for regular branching programs. *SICOMP* 43 (3), pages 973–986, 2014 (FOCS 2010)

26. S. Moran and A. Yehudayoff. A note on average-case sorting. *Order* 33, pages 23–28, 2016
27. A. Wigderson and A. Yehudayoff. Population recovery and partial identification. *Machine Learning* 102, pages 29–56, 2016 (FOCS 2012)
28. G. Kol, S. Moran, A. Shpilka and A. Yehudayoff. Direct sum fails for zero error average communication. *Algorithmica* 76 (3), pages 782–795, 2016 (ITCS 2014)
29. S. Moran and A. Yehudayoff. Sample compression schemes for VC classes. *JACM* 63 (3), pages 1– 21, 2016
30. D. Ellis, E. Friedgut, G. Kindler and A. Yehudayoff. Geometric stability via information theory. *Discrete Analysis* 10, DOI: 10.19086/da.784, 2016
31. N. Alon, S. Moran and A. Yehudayoff. Sign rank versus VC dimension. *Sbornik Mathematics* 208 (12), pages 4–41, 2017
32. A. Yehudayoff. An elementary exposition of topological overlap in the plane. *Discrete & Computational Geometry* 58, pages 255–264, 2017
33. K. Censor-Hillel, T. Kavitha, A. Paz and A. Yehudayoff. Distributed constructions of purely additive spanners. *Distributed Computing* 31, pages 223–240, 2018 (DISC 2016)
34. S. Ben-David, P. Hrubeš, S. Moran, A. Shpilka and A. Yehudayoff. Learning can be undecidable. *Nature Machine Intelligence* 1, pages 44–48, 2019 (invited to STOC 21)
35. O. Shir and A. Yehudayoff. On the covariance-Hessian relation in evolution strategies. *Theoretical Computer Science* 801, pages 157–174, 2020 (FOGA 2017)
36. A. Yehudayoff. Pointer chasing via triangular discrimination. *Combinatorics, Probability and Computing*, 29 (4), pages 485–494, 2020
37. S. Moran and A. Yehudayoff. On weak ε -nets and the Radon number. *Discrete and Computational Geometry* 64 (4), pages 1125–1140, 2020 (SoCG 2019)
38. P. Gopalan and A. Yehudayoff. Concentration for limited independence via inequalities for the elementary symmetric polynomials. *Theory of Computing* 16, pages 1–29, 2020
39. Z. Jiang and A. Yehudayoff. An isoperimetric inequality for Hamming balls and local expansion in hypercubes. *Electronic Journal of Combinatorics* 29(1), 2022
40. A. Rao and A. Yehudayoff. Anti-concentration and the exact gap-hamming problem. *SIAM J. Discret. Math.* 36(2), pages 1071–1092, 2022
41. B. E. Diamond and A. Yehudayoff. Explicit exponential lower bounds for exact hyperplane covers. *Discret. Math.* 345(11) 113114, 2022
42. G. Dank, T. Buber, G. Polliack, G. Aviram, A. Rice, A. Yehudayoff, M. S. Kent. A pilot study for a non-invasive system for detection of malignancy in canine subcutaneous and cutaneous masses using machine learning. *Frontiers in Veterinary Science* 10, 2023

43. Daniel Avraham and Amir Yehudayoff. On Blocky Ranks Of Matrices. *Comput. Complex.* 33(1), 2024
44. G. Yehuda and A. Yehudayoff. A lower bound for essential covers of the cube. *Combinatorica* 2024
45. O. Angel, J. Richey, Y. Spinka and A. Yehudayoff. Random walks on regular trees can not be slowed down. *Electronic Journal of Probability* 2024
46. E. Milman and A. Yehudayoff. A Blaschke-Santaló inequality for unconditional log-concave measures. *Annales de la Faculté des Sciences de Toulouse* 2025
47. E. Milman, S. Shabelman and A. Yehudayoff. Fixed and periodic points of the intersection body operator. *Inventiones Mathematicae* 2025
48. G. Yehuda and A. Yehudayoff. A LYM inequality for product measures. *Discret. Math.* 349(3), 2026
49. S. Gat, G. Aviram, H. Toledano, M. Tshuva, A. Gur, S. Toledano, A. Onn, G. Polliack and A. Yehudayoff. A Novel POC modality for ex vivo malignancy detection using thermal diffusion analysis: a case study of lung cancer. *Applied Sciences* 2026
50. E. Bakaev and A. Yehudayoff. A note on the affine plank conjecture. *Proc. of the AMS*, to appear

Peer-reviewed conference proceedings

51. P. Hrubeš, A. Wigderson and A. Yehudayoff. Relationless completeness and separations. *CCC* 2010
52. Z. Dvir, A. Rao, A. Wigderson and A. Yehudayoff. Restriction access. *ITCS* 2012
53. Z. Dvir, G. Malod, S. Perifel and A. Yehudayoff. Separating multilinear branching programs and formulas. *STOC* 2012
54. M. Braverman, A. Rao, O. Weinstein and A. Yehudayoff. Direct products in communication complexity. *FOCS* 2013
55. P. Hrubeš and A. Yehudayoff. Formulas are exponentially stronger than monotone circuits in non commutative setting. *CCC* 2013
56. M. Braverman, A. Rao, O. Weinstein and A. Yehudayoff. Direct product via round-preserving compression. *ICALP* 2013
57. A. Rao and A. Yehudayoff. Simplified lower bounds on the multiparty communication complexity of disjointness. *CCC* 2015
58. B. Bauer, S. Moran and A. Yehudayoff. Internal compression of protocols to entropy. *APPROX-RANDOM* 2015

59. P. Hrubeš and A. Yehudayoff. On isoperimetric profiles and computational complexity. ICALP 2016
60. O. David, S. Moran and A. Yehudayoff. On statistical learning via the lens of compression. NeurIPS oral presentation 2016
61. N. Alon, M. Babaioff, Y. A. Gonczarowski, Y. Mansour, S. Moran and A. Yehudayoff. Submultiplicative Glivenko-Cantelli and uniform convergence of revenues. NeurIPS 2017
62. R. Bassily, S. Moran, I. Nachum, J. Shafer and A. Yehudayoff. Learners that leak little information. ALT 2018
63. I. Nachum, J. Shafer and A. Yehudayoff. A direct sum result for the information complexity of learning. COLT 2018
64. D. Kane, R. Livni, S. Moran and A. Yehudayoff. On communication complexity of classification problems. COLT 2019
65. I. Haitner, N. Mazon, R. Oshman, O. Reingold and A. Yehudayoff. On the communication complexity of key-agreement protocols. ITCS 2019
66. A. Yehudayoff. Separating monotone VP and VNP. STOC 2019
67. P. Hrubeš, S. N. Ramamoorthy, A. Rao and A. Yehudayoff. Lower bounds on balancing sets and depth-2 threshold circuits. ICALP 2019
68. I. Nachum and A. Yehudayoff. Average-case information complexity of learning. ALT 2019
69. S. Gocht, J. Nordstrom and A. Yehudayoff. On division versus saturation in pseudo-boolean solving. IJCAI 2019
70. I. Nachum and A. Yehudayoff. On symmetry and initialization for neural networks. Latin American Symposium on Theoretical Informatics 2020
71. S. Goldwasser, G. Rothblum, J. Shafer and A. Yehudayoff. Interactive proofs for verifying machine learning. ITCS 2021
72. O. Bousquet, S. Hanneke, S. Moran, R. van Handel and A. Yehudayoff. A theory of universal learning. STOC 2021
73. P. Hrubeš and A. Yehudayoff. Shadows of Newton Polytopes. CCC 2021
74. N. Brukhim, D. Carmon, I. Dinur, S. Moran and A. Yehudayoff. A characterization of multiclass learnability. FOCS 2022
75. Z. Chase, S. Moran and A. Yehudayoff. Stability and Replicability in Learning. FOCS 2023
76. D. Carmon, R. Livni and A. Yehudayoff. The Sample Complexity Of ERM's In Stochastic Convex Optimization. AISTAT 24

77. Z. Chase, B. Chornomaz, S. Moran and A. Yehudayoff. Local Borsuk-Ulam, stability, and replicability. STOC 2024
78. Z. Chase, B. Chornomaz, S. Hanneke, S. Moran and A. Yehudayoff. Dual VC dimension obstructs sample compression by embeddings. COLT 2024
79. N. Alon, S. Moran, H. Schefler and A. Yehudayoff. A unified characterization of private learnability via graph theory. COLT 2024
80. S. Hanneke, S. Moran, A. Shlimovich and A. Yehudayoff. Data Selection for ERM's. COLT 2025
81. I. Orzel, S. Srinivasan, S. Tavena and A. Yehudayoff. The algebraic cost of a boolean sum. FSTTCS 2025
82. E. Bakaev, F. Brunck, C. Hertrich, J. Stade and A. Yehudayoff. Better neural network expressivity: subdividing the simplex. STOC 2026
83. B. Cavalar, T. Fabris, P. Mukhopadhyay, S. Srinivasan and A. Yehudayoff. Negations are powerful even in small depth. STOC 2026
84. T. Fabris, N. Limaye, S. Srinivasan and A. Yehudayoff. Multilinear algebraic branching programs and the min-partition rank method. CCC 2026

Surveys

85. A. Shpilka and A. Yehudayoff. Arithmetic circuits: a survey of recent results and open questions. Foundations and Trends in Theoretical Computer Science 5 (1), pages 1–72, 2011
86. A. Yehudayoff. Proving expansion in three steps. SIGACT News Complexity Theory Column, September 2012

Manuscripts

87. G. Yehuda and A. Yehudayoff. Slicing the hypercube is not easy. arXiv
88. S. Iyer, A. Rao, V. Reis, T. Rothvoss and A. Yehudayoff. Tight bounds on the Fourier growth of bounded functions on the hypercube. arXiv
89. S. Srinivasan and A. Yehudayoff. The discrepancy of greater-than. arXiv
90. J. Andersson and A. Yehudayoff. On the space complexity of online convolution. arXiv